Dermascan AI - Viva Questions & Answers

General Project Questions

Q: What is the main objective of your project?
A: To provide a quick, accessible, and accurate preliminary diagnosis of skin diseases using AI, and to connect patients with dermatologists for professional consultation.

Q: What problem are you solving?
A: Shortage of dermatologists, long waiting times for appointments, and the high cost of initial consultations. Our app acts as a first line of triage.

Q: What is the scope of this project?
A: It covers 8 specific skin diseases. It is a web-based application serving Patients, Doctors, and Administrators.

Technical Questions (Backend & Frontend)

Q: Which framework did you use and why?
A: We used **Django** (Python) because it follows the MVT (Model-View-Template) architecture, has built-in security features (like CSRF protection), and comes with a powerful Admin panel out of the box.

Q: What database are you using?
A: We are using **MySQL**. It is a robust, relational database management system that is widely used in production environments for its reliability and scalability.

Q: How does the "Find Specific Doctor" feature work?
A: We query the database for doctors who match the specific specialization required for the predicted disease (taken from the ML result) and display those profiles to the user.

Q: How did you implement user authentication?
A: We used Django's built-in `auth` system, which handles session management, password hashing (PBKDF2), and user permissions securely.

AI & Machine Learning Questions

Q: Which algorithm did you use for the skin disease detection?
A: We used **MobileNetV2**, a Convolutional Neural Network (CNN) architecture optimized for speed and efficiency.

Q: Why MobileNetV2 and not ResNet or VGG?
A: MobileNetV2 is much smaller and faster (fewer parameters), making it ideal for web deployment

where we need quick predictions without heavy server load.

Q: What is the accuracy of your model?
A: The model achieves an accuracy of approximately **94%** on the validation set.

Q: Where did you get the dataset?
A: We curated a dataset known as "Dermnet" and other open-source medical repositories, containing thousands of images across the 8 classes.

Q: How do you handle images of different sizes?
A: All user-uploaded images are resized to **224x224 pixels** and normalized (pixel values divided by 255) before being fed into the model.

Future Scope & Challenges

Q: What are the limitations of your project?
A: It currently only detects 8 specific diseases. It is not a replacement for a real doctor, only a decision support tool. Lighting conditions in user photos can affect accuracy.

Q: What would you add if you had more time?
A: A mobile app (React Native), real-time video consultation integration, and a more extensive dataset covering more skin conditions.

Q: What was the biggest challenge?
A: Integrating the Deep Learning model with the Django backend and ensuring the image preprocessing pipeline matched exactly what was used during training.